

Zee — Movie Recommender System

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Problem Statement

Zee (an OTT / streaming platform) wants to improve user experience and engagement by showing each user personalized movie recommendations instead of generic, one-size-fits-all suggestions. The platform has historical data of movie ratings given by thousands of users, along with user demographic details and movie metadata.

The core problem is that with thousands of movies available, users struggle to discover content they will actually enjoy. This leads to poor engagement, more time spent searching, and a higher chance that users leave the platform.

The goal is to leverage the ratings given by a user — and by other users who are *similar* to them — to predict what movies a given user is most likely to enjoy, and recommend those movies to them.

This is a classic **Collaborative Filtering** based **Recommender System** problem.

Objective

The objective of this case study is to design and build a recommender system that suggests personalized movie recommendations. Specifically, we aim to:

1. Read, format and merge the three data files (ratings, users, movies) into a single workable dataframe.
2. Perform Exploratory Data Analysis

Loading the Data

We mount Google Drive and read the three `::`-separated data files.

```
In [ ]: from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
In [ ]: import os
```

```
PATH = '/content/'
print(os.listdir(PATH))
```

```
['.config', 'zee-ratings.dat', 'drive', 'zee-users.dat', 'zee-movies.dat', 'sample_data']
```

Reading and Merging Files

The files use `::` as a separator. We give proper column names while reading, then merge ratings + users + movies into one consolidated dataframe.

```
In [ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

ratings = pd.read_csv(PATH + 'zee-ratings.dat', sep='::', engine='python',
                      names=['UserID', 'MovieID', 'Rating', 'Timestamp'],
                      encoding='ISO-8859-1', skiprows=1)

users = pd.read_csv(PATH + 'zee-users.dat', sep='::', engine='python',
                   names=['UserID', 'Gender', 'Age', 'Occupation', 'Zip-code'],
                   encoding='ISO-8859-1', skiprows=1)

movies = pd.read_csv(PATH + 'zee-movies.dat', sep='::', engine='python',
                    names=['MovieID', 'Title', 'Genres'],
                    encoding='ISO-8859-1', skiprows=1)

print("Ratings:", ratings.shape)
print("Users  :", users.shape)
```

```
print("Movies :", movies.shape)
ratings.head()
```

Ratings: (1000209, 4)
 Users : (6040, 5)
 Movies : (3883, 3)

```
Out[ ]:   UserID  MovieID  Rating  Timestamp
0        1      1193        5   978300760
1        1        661        3   978302109
2        1        914        3   978301968
3        1      3408        4   978300275
4        1      2355        5   978824291
```

display(users.head()) display(movies.head())

```
In [ ]: df = ratings.merge(users, on='UserID').merge(movies, on='MovieID')
print("Merged dataframe shape:", df.shape)
df.head()
```

Merged dataframe shape: (1000209, 10)

```
Out[ ]:   UserID  MovieID  Rating  Timestamp  Gender  Age  Occupation  Zip-
code  Title  Genres
0        1      1193        5   978300760        F    1         10   48067  One Flew Over the Cuckoo's
Nest (1975)        Drama
1        1        661        3   978302109        F    1         10   48067  James and the Giant Peach
(1996)  Animation|Children's|Musical
2        1        914        3   978301968        F    1         10   48067    My Fair Lady (1964)        Musical|Romance
3        1      3408        4   978300275        F    1         10   48067    Erin Brockovich (2000)        Drama
4        1      2355        5   978824291        F    1         10   48067    Bug's Life, A (1998)  Animation|Children's|Comedy
```

EDA, Data Cleaning & Feature Engineering

We review the structure of the dataset, check for missing values and duplicates, perform type conversions, map the coded columns (Age, Occupation) to readable labels, and derive a new feature `ReleaseYear` (and `Decade`) from the movie title.

```
In [ ]: df.info()
print("\nMissing values:\n", df.isnull().sum())
print("\nDuplicate rows:", df.duplicated().sum())
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000209 entries, 0 to 1000208
Data columns (total 10 columns):
#   Column          Non-Null Count  Dtype
---  -
0   UserID          1000209 non-null  object
1   MovieID         1000209 non-null  object
2   Rating          1000209 non-null  object
3   Timestamp       1000209 non-null  object
4   Gender          1000209 non-null  object
5   Age            1000209 non-null  object
6   Occupation      1000209 non-null  object
7   Zip-code       1000209 non-null  object
8   Title          1000209 non-null  object
9   Genres          1000209 non-null  object
dtypes: object(10)
memory usage: 76.3+ MB
```

Missing values:

```
UserID          0
MovieID         0
Rating          0
Timestamp       0
Gender          0
Age            0
Occupation      0
Zip-code       0
Title          0
Genres         0
dtype: int64
```

Duplicate rows: 0

Out[]:

	UserID	MovielD	Rating	Timestamp	Gender	Age	Occupation	Zip-code	Title	Genres
count	1000209	1000209	1000209	1000209	1000209	1000209	1000209	1000209	1000209	1000209
unique	6040	3706	5	458455	2	7	21	3439	3706	301
top	4169	2858	4	975528402	M	25	4	94110	American Beauty (1999)	Comedy
freq	2314	3428	348971	30	753769	395556	131032	3802	3428	116883

In []:

```
# Convert Age column to integer type as it was likely parsed as object (string)
df['Age'] = df['Age'].astype(int)

# Map Age codes -> readable groups
age_map = {1:"Under 18",18:"18-24",25:"25-34",35:"35-44",
           45:"45-49",50:"50-55",56:"56+"}
df['Age_group'] = df['Age'].map(age_map)

# Convert Occupation column to integer type
df['Occupation'] = df['Occupation'].astype(int)

# Map Occupation codes -> Labels
occ_map = {0:"other",1:"academic/educator",2:"artist",3:"clerical/admin",
           4:"college/grad student",5:"customer service",6:"doctor/health care",
           7:"executive/managerial",8:"farmer",9:"homemaker",10:"K-12 student",
           11:"lawyer",12:"programmer",13:"retired",14:"sales/marketing",
           15:"scientist",16:"self-employed",17:"technician/engineer",
           18:"tradesman/craftsman",19:"unemployed",20:"writer"}
df['Occupation_label'] = df['Occupation'].map(occ_map)

# Derive Release Year from Title e.g. "Toy Story (1995)"
df['ReleaseYear'] = df['Title'].str.extract(r'\((\d{4})\)').astype(float)
df['Decade'] = (df['ReleaseYear']//10*10)

# Convert timestamp to datetime (optional, useful feature)
df['RatingTime'] = pd.to_datetime(df['Timestamp'], unit='s')

df[['Title','ReleaseYear','Decade','Age_group','Occupation_label']].head()
```

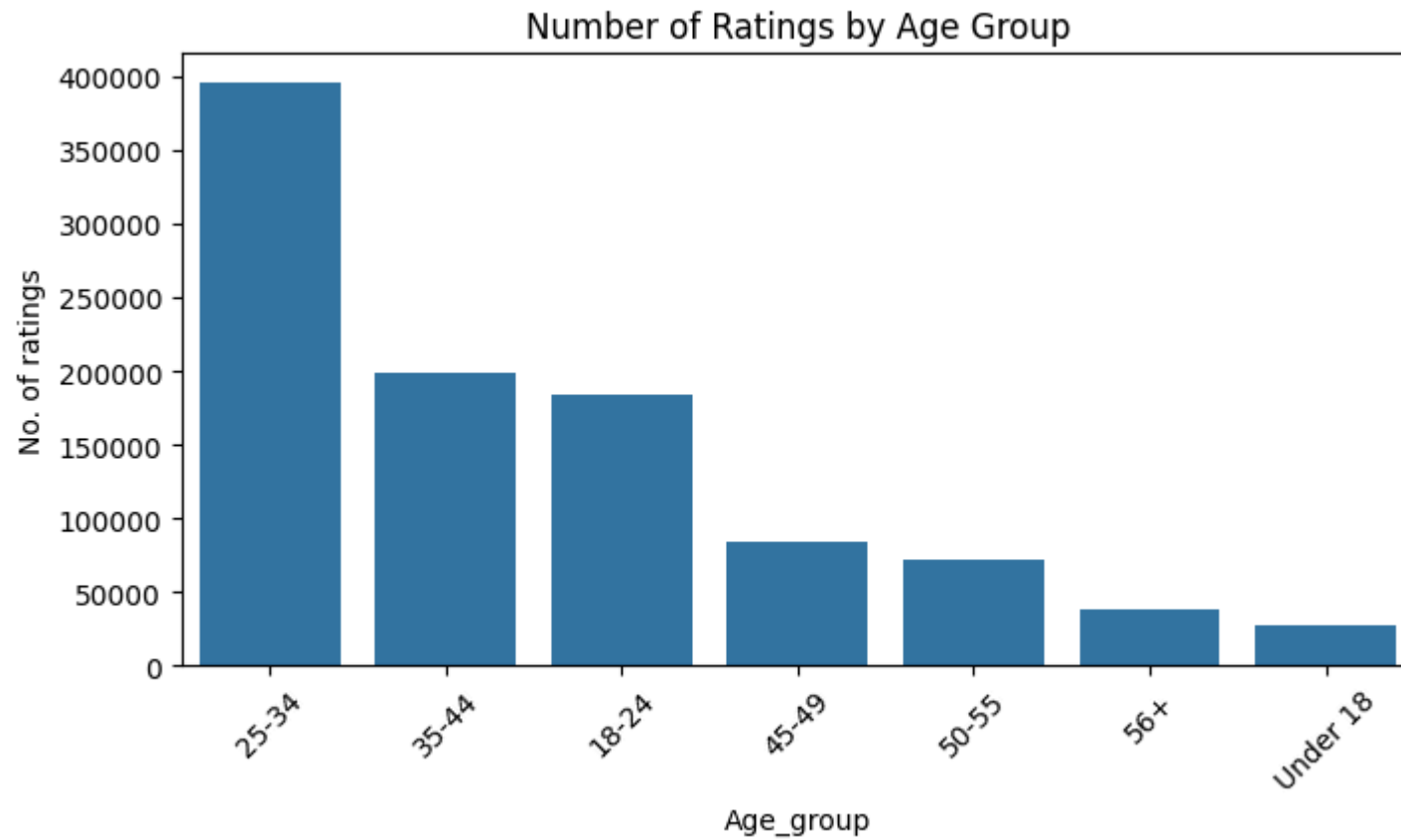
Out[]:

	Title	ReleaseYear	Decade	Age_group	Occupation_label
0	One Flew Over the Cuckoo's Nest (1975)	1975.0	1970.0	Under 18	K-12 student
1	James and the Giant Peach (1996)	1996.0	1990.0	Under 18	K-12 student
2	My Fair Lady (1964)	1964.0	1960.0	Under 18	K-12 student
3	Erin Brockovich (2000)	2000.0	2000.0	Under 18	K-12 student
4	Bug's Life, A (1998)	1998.0	1990.0	Under 18	K-12 student

```
In [ ]: # Convert Age column to integer type as it was likely parsed as object (string)
df['Age'] = df['Age'].astype(int)

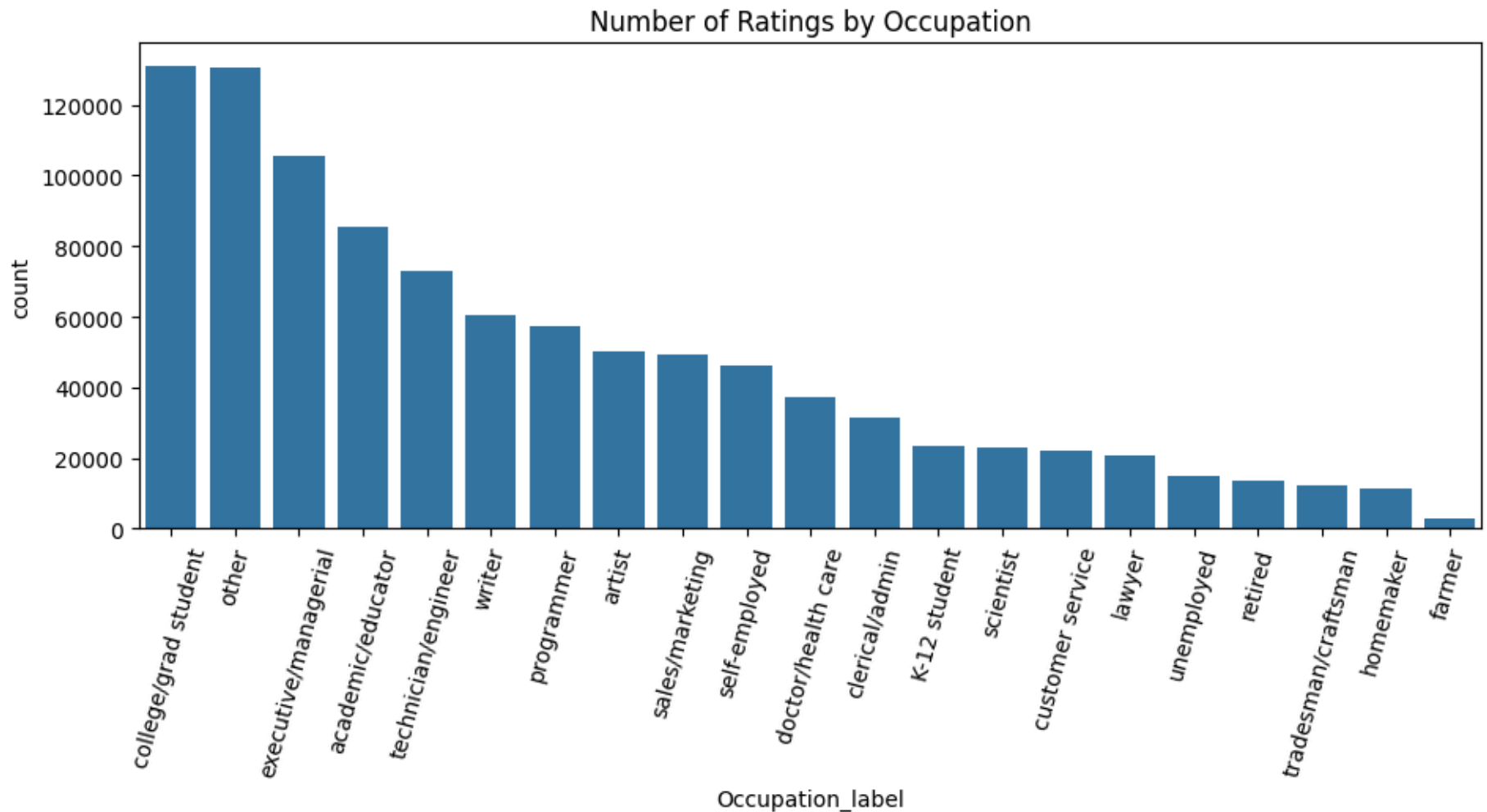
# Map Age codes -> readable groups (re-applying the mapping as Age_group was previously all NaN)
age_map = {1:"Under 18",18:"18-24",25:"25-34",35:"35-44",
           45:"45-49",50:"50-55",56:"56+"}
df['Age_group'] = df['Age'].map(age_map)

plt.figure(figsize=(8,4))
order = df['Age_group'].value_counts().index
sns.countplot(data=df, x='Age_group', order=order)
plt.title('Number of Ratings by Age Group'); plt.xticks(rotation=45); plt.ylabel('No. of ratings')
plt.show()
print("Q1 Answer – Most active age group:", df['Age_group'].value_counts().idxmax())
```



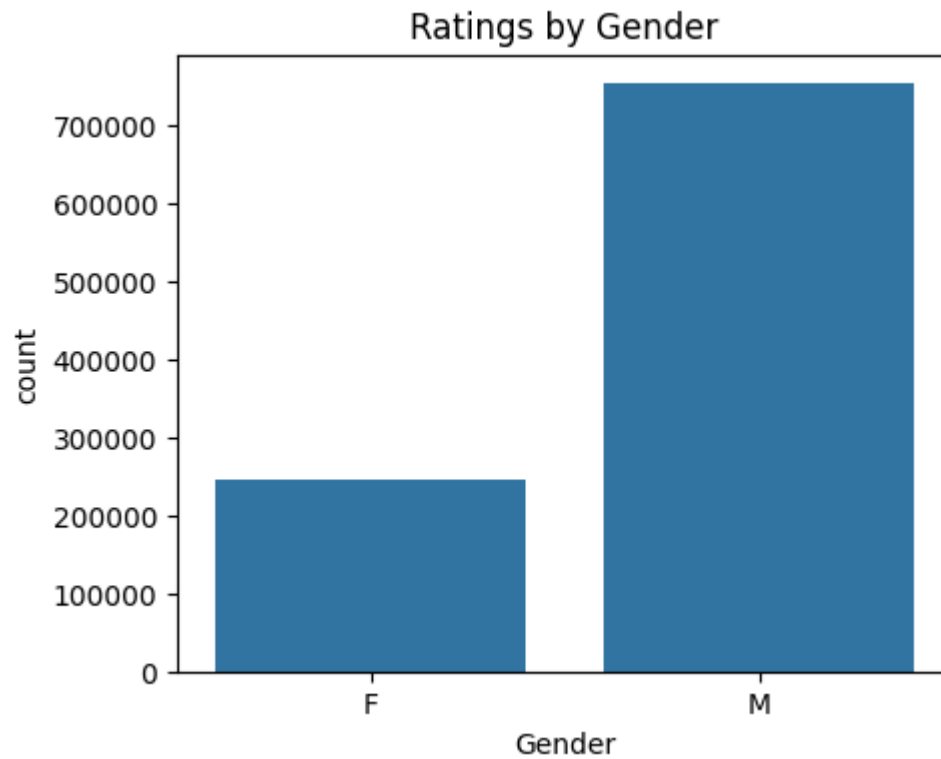
Q1 Answer – Most active age group: 25-34

```
In [ ]: plt.figure(figsize=(11,4))
order = df['Occupation_label'].value_counts().index
sns.countplot(data=df, x='Occupation_label', order=order)
plt.title('Number of Ratings by Occupation'); plt.xticks(rotation=75)
plt.show()
print("Q2 Answer – Most active profession:", df['Occupation_label'].value_counts().idxmax())
```

Q2 Answer – Most active profession: college/grad student

```
In [ ]: plt.figure(figsize=(5,4))
sns.countplot(data=df, x='Gender')
plt.title('Ratings by Gender'); plt.show()
print("Gender wise rating counts:\n", df['Gender'].value_counts())
print("\nQ3 – Most raters are Male?", df['Gender'].value_counts().idxmax()=='M')
```



Gender wise rating counts:

Gender

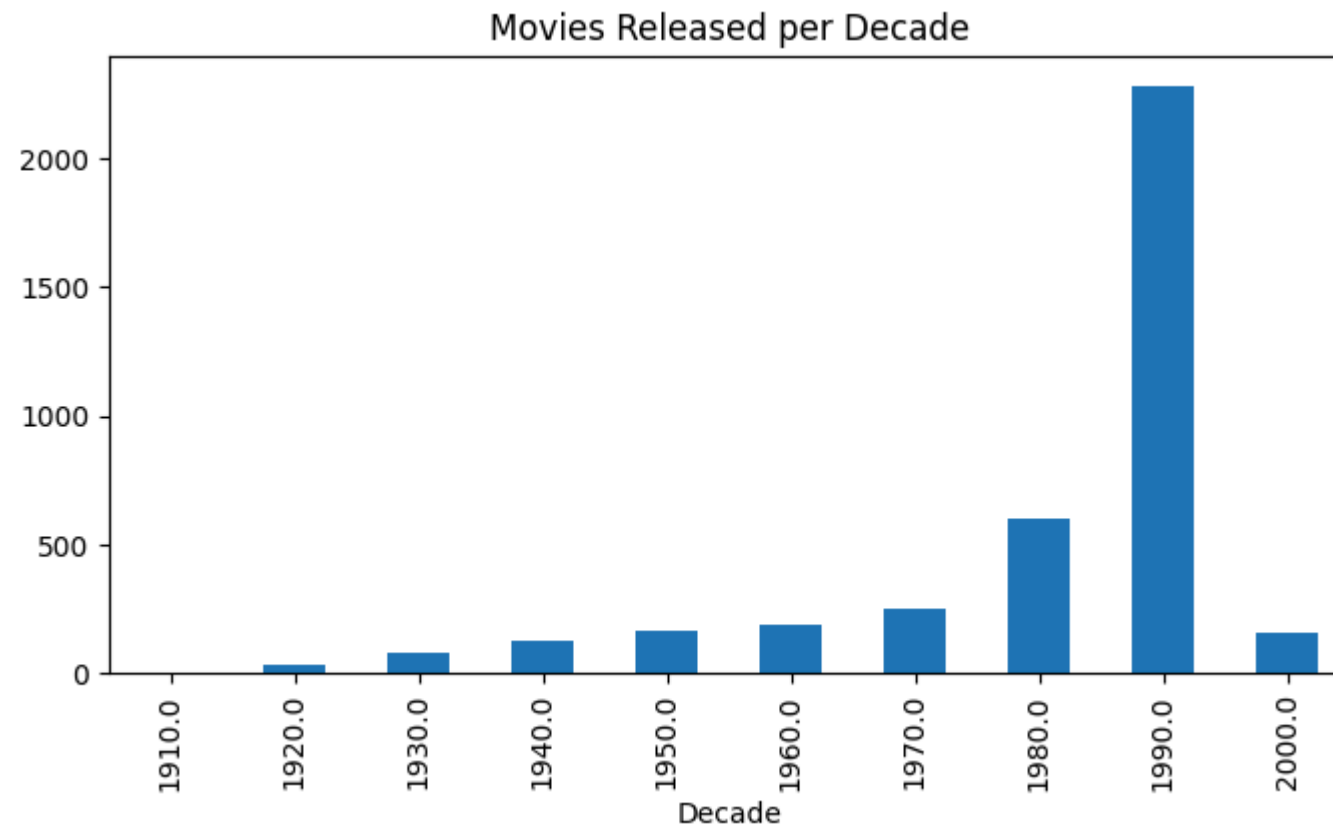
M 753769

F 246440

Name: count, dtype: int64

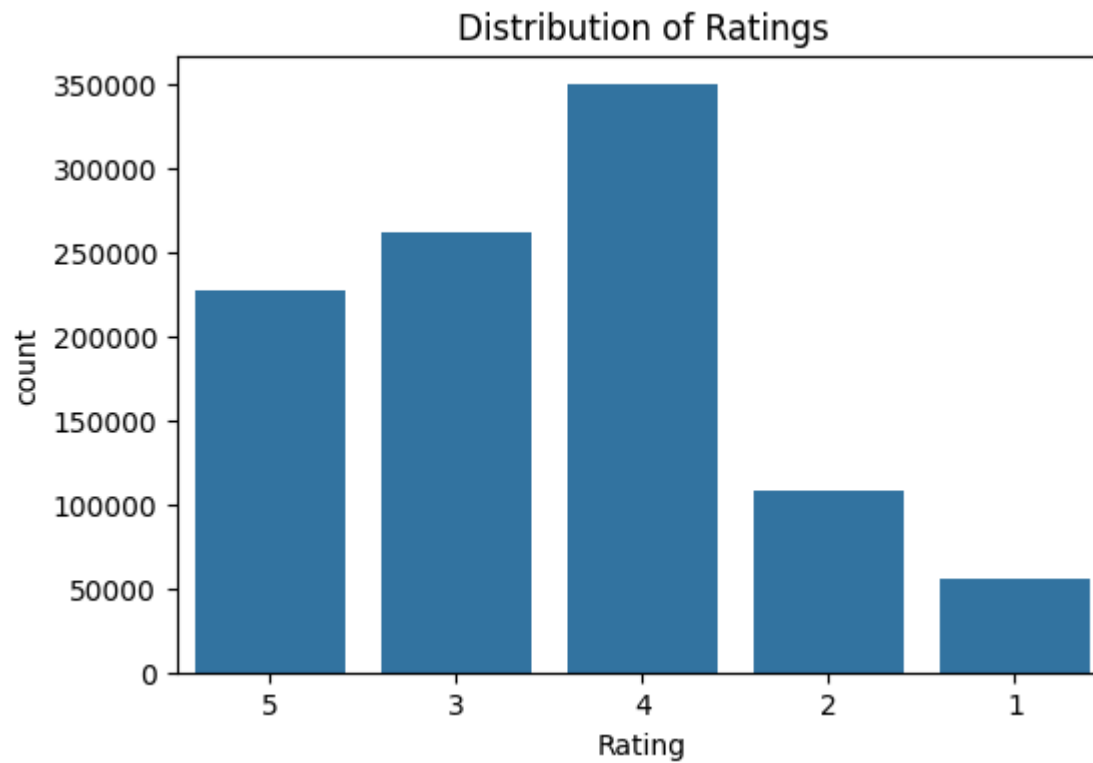
Q3 – Most raters are Male? True

```
In [ ]: plt.figure(figsize=(8,4))
# unique movies only (taaki duplicate count na ho)
mv = movies.copy()
mv['Year'] = mv['Title'].str.extract(r'\((\d{4})\)').astype(float)
mv['Decade'] = (mv['Year']//10*10)
mv['Decade'].value_counts().sort_index().plot(kind='bar')
plt.title('Movies Released per Decade'); plt.xlabel('Decade'); plt.show()
print("Q4 Answer – Decade with most movies:", int(mv['Decade'].value_counts().idxmax()), "s")
```



Q4 Answer – Decade with most movies: 1990 s

```
In [ ]: plt.figure(figsize=(6,4))
sns.countplot(data=df, x='Rating')
plt.title('Distribution of Ratings'); plt.show()
```



```
In [ ]: df['Rating'] = df['Rating'].astype(int)
movie_stats = df.groupby('Title').agg(
    Avg_Rating=('Rating', 'mean'),
    Num_Ratings=('Rating', 'count')
).sort_values('Num_Ratings', ascending=False)

print("Q5 Answer – Movie with maximum number of ratings:", movie_stats.index[0])
movie_stats.head(10)
```

Q5 Answer – Movie with maximum number of ratings: American Beauty (1999)

Out[]:

	Avg_Rating	Num_Ratings
Title		
American Beauty (1999)	4.317386	3428
Star Wars: Episode IV - A New Hope (1977)	4.453694	2991
Star Wars: Episode V - The Empire Strikes Back (1980)	4.292977	2990
Star Wars: Episode VI - Return of the Jedi (1983)	4.022893	2883
Jurassic Park (1993)	3.763847	2672
Saving Private Ryan (1998)	4.337354	2653
Terminator 2: Judgment Day (1991)	4.058513	2649
Matrix, The (1999)	4.315830	2590
Back to the Future (1985)	3.990321	2583
Silence of the Lambs, The (1991)	4.351823	2578

```
In [ ]: # Sirf un movies pe focus jinke paas kaafi ratings hain (e.g. >100)
popular = movie_stats[movie_stats['Num_Ratings'] > 100]
print("Top 10 highest rated movies (with >100 ratings):")
popular.sort_values('Avg_Rating', ascending=False).head(10)
```

Top 10 highest rated movies (with >100 ratings):

Out[]:

	Avg_Rating	Num_Ratings
Title		
Seven Samurai (The Magnificent Seven) (Shichinin no samurai) (1954)	4.560510	628
Shawshank Redemption, The (1994)	4.554558	2227
Godfather, The (1972)	4.524966	2223
Close Shave, A (1995)	4.520548	657
Usual Suspects, The (1995)	4.517106	1783
Schindler's List (1993)	4.510417	2304
Wrong Trousers, The (1993)	4.507937	882
Sunset Blvd. (a.k.a. Sunset Boulevard) (1950)	4.491489	470
Raiders of the Lost Ark (1981)	4.477725	2514
Rear Window (1954)	4.476190	1050

Recommender System using Pearson Correlation (Item-based)

We create a pivot table (matrix) of **users (rows) × movie titles (columns)** where each cell is the rating. Most cells will be empty (NaN) because no user rates every movie — this is a sparse matrix.

To recommend movies similar to a given movie, we compute the **Pearson Correlation** between that movie's rating column and every other movie's rating column. Movies whose ratings move together (high positive correlation) are considered "similar". We also filter out movies with too few ratings to avoid unreliable correlations.

```
In [ ]: # Users x Movies pivot table of ratings
movie_matrix = df.pivot_table(index='UserID', columns='Title', values='Rating')
print("Pivot table shape:", movie_matrix.shape)
movie_matrix.iloc[:5, :5]
```

Pivot table shape: (6040, 3706)

Out[]: **Title** **\$1,000,000 Duck (1971)** **'Night Mother (1986)** **'Til There Was You (1997)** **'burbs, The (1989)** **...And Justice for All (1979)**

UserID					
1	NaN	NaN	NaN	NaN	NaN
10	NaN	NaN	NaN	4.0	NaN
100	NaN	NaN	NaN	NaN	NaN
1000	NaN	NaN	NaN	NaN	NaN
1001	NaN	NaN	NaN	NaN	NaN

```
In [ ]: def recommend_pearson(movie_name, min_ratings=100, top_n=5):
        """Item-based recommender using Pearson correlation."""
        if movie_name not in movie_matrix.columns:
            print("Movie not found. Check exact title (with year), e.g. 'Liar Liar (1997)')")
            return None

        target_ratings = movie_matrix[movie_name]

        # Correlation of target movie with all other movies
        similar = movie_matrix.corrwith(target_ratings)
        corr_df = pd.DataFrame(similar, columns=['Correlation']).dropna()

        # Add number of ratings to filter out unreliable ones
        corr_df = corr_df.join(movie_stats['Num_Ratings'])

        # Keep only movies with enough ratings, sort by correlation
        result = corr_df[corr_df['Num_Ratings'] > min_ratings] \
            .sort_values('Correlation', ascending=False)

        # Drop the movie itself (top row) and return next top_n
        return result.drop(index=movie_name, errors='ignore').head(top_n)
```

```
In [ ]: # User input Le sakte hain:
        # movie = input("Enter a movie name (with year, e.g. 'Liar Liar (1997)'): ")

        movie = "Liar Liar (1997)"    # demonstration ke liye fixed
```

```
print(f"Top 5 movies similar to '{movie}':\n")
recommend_pearson(movie, min_ratings=100, top_n=5)
```

Top 5 movies similar to 'Liar Liar (1997)':

Out[]:

	Correlation	Num_Ratings
Title		
Life (1999)	0.576081	102
East-West (Est-ouest) (1999)	0.562221	103
Oliver & Company (1988)	0.550504	144
Jules and Jim (Jules et Jim) (1961)	0.537842	148
Solaris (Solyaris) (1972)	0.537645	143

```
In [ ]: print("Q6 Answer – Top 3 movies similar to 'Liar Liar (1997)':")
recommend_pearson("Liar Liar (1997)", min_ratings=100, top_n=3)
```

Q6 Answer – Top 3 movies similar to 'Liar Liar (1997)':

Out[]:

	Correlation	Num_Ratings
Title		
Life (1999)	0.576081	102
East-West (Est-ouest) (1999)	0.562221	103
Oliver & Company (1988)	0.550504	144

```
In [ ]: movie = "Star Wars: Episode IV - A New Hope (1977)"
print(f"Top 5 movies similar to '{movie}':\n")
recommend_pearson(movie, min_ratings=100, top_n=5)
```

Top 5 movies similar to 'Star Wars: Episode IV - A New Hope (1977)':

Out[]:

	Correlation	Num_Ratings
Title		
Star Wars: Episode V - The Empire Strikes Back (1980)	0.661552	2990
Star Wars: Episode VI - Return of the Jedi (1983)	0.574808	2883
Raiders of the Lost Ark (1981)	0.421425	2514
Dracula (1958)	0.398710	102
Girlfight (2000)	0.365521	127

```
In [ ]: [t for t in movie_matrix.columns if 'Liar Liar' in t]
```

Out[]: ['Liar Liar (1997)']

Recommender System using Cosine Similarity (Nearest Neighbors)

Cosine similarity measures the angle between two rating vectors (ranges 0 to 1 for non-negative ratings; in general -1 to 1). Unlike Pearson, it does not subtract the mean.

Steps:

1. Build user×movie and movie×user matrices, fill NaN with 0.
2. Display a sample **user-user** and **item-item** similarity matrix (for demonstration).
3. Convert the sparse matrix to a **CSR matrix** for efficiency.
4. Use **KNN (NearestNeighbors)** with cosine metric to find 5 movies similar to a given movie.

```
In [ ]: from sklearn.metrics.pairwise import cosine_similarity

# Fill NaN with 0 on the pivot table
mat_filled = movie_matrix.fillna(0)

# ---- USER-USER similarity (demo on a small subset to keep it readable) ----
```

```

user_sample = mat_filled.iloc[:5]          # first 5 users
user_sim = cosine_similarity(user_sample)
user_sim_df = pd.DataFrame(user_sim, index=user_sample.index, columns=user_sample.index)
print("User-User Similarity Matrix (sample of 5 users):")
display(user_sim_df.round(3))

# ---- ITEM-ITEM similarity (demo on a small subset) ----
item_sample = mat_filled.iloc[:, :5]      # first 5 movies
item_sim = cosine_similarity(item_sample.T)
item_sim_df = pd.DataFrame(item_sim, index=item_sample.columns, columns=item_sample.columns)
print("\nItem-Item Similarity Matrix (sample of 5 movies):")
display(item_sim_df.round(3))

```

User-User Similarity Matrix (sample of 5 users):

UserID 1 10 100 1000 1001

UserID

1	1.000	0.255	0.124	0.208	0.139
10	0.255	1.000	0.258	0.279	0.155
100	0.124	0.258	1.000	0.298	0.076
1000	0.208	0.279	0.298	1.000	0.095
1001	0.139	0.155	0.076	0.095	1.000

Item-Item Similarity Matrix (sample of 5 movies):

Title	\$1,000,000 Duck (1971)	'Night Mother (1986)	'Til There Was You (1997)	'burbs, The (1989)	...And Justice for All (1979)
Title					
\$1,000,000 Duck (1971)	1.000	0.072	0.037	0.079	0.061
'Night Mother (1986)	0.072	1.000	0.115	0.116	0.160
'Til There Was You (1997)	0.037	0.115	1.000	0.099	0.066
'burbs, The (1989)	0.079	0.116	0.099	1.000	0.144
...And Justice for All (1979)	0.061	0.160	0.066	0.144	1.000

```
In [ ]: from scipy.sparse import csr_matrix
        from sklearn.neighbors import NearestNeighbors

        # movie x user matrix (rows = movies)
        movie_user_mat = movie_matrix.fillna(0).T      # Title as rows, UserID as columns

        # Optional: filter to popular movies for cleaner recommendations
        popular_titles = movie_stats[movie_stats['Num_Ratings'] > 50].index
        movie_user_mat = movie_user_mat.loc[movie_user_mat.index.intersection(popular_titles)]

        # CSR sparse matrix
        csr = csr_matrix(movie_user_mat.values)
        print("CSR matrix shape:", csr.shape)

        # KNN model with cosine similarity
        knn = NearestNeighbors(metric='cosine', algorithm='brute', n_neighbors=6)
        knn.fit(csr)
```

CSR matrix shape: (2499, 6040)

Out[]:

```
NearestNeighbors
NearestNeighbors(algorithm='brute', metric='cosine', n_neighbors=6)
```

In []:

```
def recommend_knn(movie_name, n=5):
    """Recommend n similar movies using KNN + cosine similarity."""
    if movie_name not in movie_user_mat.index:
        print("Movie not found in matrix. Try exact title with year.")
        return None

    idx = movie_user_mat.index.get_loc(movie_name)
    distances, indices = knn.kneighbors(csr[idx], n_neighbors=n+1)

    print(f"Movies similar to '{movie_name}':\n")
    recs = []
    for dist, i in zip(distances.flatten()[1:], indices.flatten()[1:]): # skip itself
        title = movie_user_mat.index[i]
        recs.append((title, round(1 - dist, 3))) # cosine similarity = 1 - distance
    return pd.DataFrame(recs, columns=['Movie', 'Cosine_Similarity'])
```

In []:

```
# movie = input("Enter a movie name (with year): ")
movie = "Liar Liar (1997)"
recommend_knn(movie, n=5)
```

Movies similar to 'Liar Liar (1997)':

Out[]:

	Movie	Cosine_Similarity
0	Mrs. Doubtfire (1993)	0.557
1	Ace Ventura: Pet Detective (1994)	0.517
2	Dumb & Dumber (1994)	0.513
3	Home Alone (1990)	0.511
4	Wayne's World (1992)	0.499

```
In [ ]: recommend_knn("Toy Story (1995)", n=5)
```

Movies similar to 'Toy Story (1995)':

```
Out[ ]:
```

	Movie	Cosine_Similarity
0	Toy Story 2 (1999)	0.633
1	Groundhog Day (1993)	0.611
2	Aladdin (1992)	0.606
3	Bug's Life, A (1998)	0.579
4	Back to the Future (1985)	0.570

```
In [ ]: [t for t in movie_user_mat.index if 'Toy Story' in t]
```

```
Out[ ]: ['Toy Story (1995)', 'Toy Story 2 (1999)']
```

Recommender System using Matrix Factorization (SVD)

Matrix Factorization decomposes the large sparse user×item rating matrix into two smaller matrices of latent factors (embeddings) — one for users and one for items — of dimension `d`. Predicted rating = dot product of the user and item embedding vectors.

We use the **Surprise** library's SVD with `n_factors = 4` ($d = 4$). We do a train-test split, train the model, and evaluate using **RMSE** and **MAPE**.

```
In [ ]: !pip install scikit-surprise -q
```

```
0.0/3.1 MB ? eta -:--:--
0.3/3.1 MB 9.0 MB/s eta 0:00:01
3.0/3.1 MB 42.9 MB/s eta 0:00:01
3.1/3.1 MB 29.4 MB/s eta 0:00:00
```

```
In [ ]: from surprise import Dataset, Reader, SVD
from surprise.model_selection import train_test_split
```

```

from surprise import accuracy
import numpy as np

# Surprise expects: user, item, rating
reader = Reader(rating_scale=(1, 5))
data = Dataset.load_from_df(ratings[['UserID', 'MovieID', 'Rating']], reader)

# Train-test split (Bonus question: yahi MF ka train-test split hai)
trainset, testset = train_test_split(data, test_size=0.2, random_state=42)

# Matrix Factorization with d = 4 latent factors
svd = SVD(n_factors=4, random_state=42)
svd.fit(trainset)
predictions = svd.test(testset)
print("Model trained. Number of test predictions:", len(predictions))

```

Model trained. Number of test predictions: 200042

```

In [ ]: # RMSE
rmse = accuracy.rmse(predictions, verbose=True)

# MAPE (manually compute)
y_true = np.array([pred.r_ui for pred in predictions])
y_pred = np.array([pred.est for pred in predictions])
mape = np.mean(np.abs((y_true - y_pred) / y_true)) * 100

print(f"\nRMSE: {rmse:.4f}")
print(f"MAPE: {mape:.2f}%")
print("\nQ10 Answer – RMSE and MAPE above (note these values).")

```

RMSE: 0.8830

RMSE: 0.8830

MAPE: 26.95%

Q10 Answer – RMSE and MAPE above (note these values).

```

In [ ]: # Example: User 1 ke Liye Movie 1193 ki predicted rating
pred = svd.predict(uid=1, iid=1193)
print(pred)

```

```
user: 1          item: 1193      r_ui = None    est = 4.57    {'was_impossible': False}
```

Using MF Embeddings for Similarity

After training, the model gives us latent embedding vectors for each user (`pu`) and each item (`qi`). We redesign our similarity functions to use these 4-dimensional embeddings (instead of raw rating vectors) and find similar items/users via cosine similarity.

```
In [ ]: from sklearn.metrics.pairwise import cosine_similarity

# Item embeddings (qi) and user embeddings (pu)
item_factors = svd.qi    # shape: (n_items, 4)
user_factors = svd.pu    # shape: (n_users, 4)

# Maps: raw id -> inner id used by surprise
movie_id_to_title = movies.set_index('MovieID')['Title'].to_dict()

def item_embedding(movie_id):
    inner = trainset.to_inner_id(movie_id)
    return item_factors[inner]

def similar_items_mf(movie_id, top_n=5):
    """Find similar movies using MF item embeddings (d=4)."""
    try:
        target = item_embedding(movie_id).reshape(1, -1)
    except ValueError:
        print("Movie not in training set."); return None
    sims = cosine_similarity(target, item_factors).flatten()

    # Build list of (raw movie id, similarity)
    results = []
    for inner_id, score in enumerate(sims):
        raw_id = trainset.to_raw_id(inner_id)
        results.append((movie_id_to_title.get(raw_id, raw_id), score))
    res_df = pd.DataFrame(results, columns=['Movie', 'Similarity']) \
        .sort_values('Similarity', ascending=False)
    return res_df.iloc[1:top_n+1]    # skip itself
```

```
In [ ]: # Example: Liar Liar ka MovieID dhoondhein
liar_id = movies[movies['Title'].str.contains('Liar Liar', na=False)][ 'MovieID' ].values[0]
print("Movies similar to 'Liar Liar' using MF embeddings:")
similar_items_mf(liar_id, top_n=5)
```

Movies similar to 'Liar Liar' using MF embeddings:

```
Out[ ]:
```

	Movie	Similarity
864	Where Eagles Dare (1969)	0.998941
133	Basic Instinct (1992)	0.996225
151	Indiana Jones and the Last Crusade (1989)	0.993393
1126	Romancing the Stone (1984)	0.993008
1877	Dangerous Beauty (1998)	0.992786

```
In [ ]: def similar_users_mf(user_id, top_n=5):
        """Find similar users using MF user embeddings (d=4)."""
        try:
            inner = trainset.to_inner_uid(user_id)
        except ValueError:
            print("User not in training set."); return None
        target = user_factors[inner].reshape(1, -1)
        sims = cosine_similarity(target, user_factors).flatten()

        results = []
        for inner_id, score in enumerate(sims):
            raw_id = trainset.to_raw_uid(inner_id)
            results.append((raw_id, score))
        res_df = pd.DataFrame(results, columns=['UserID', 'Similarity']) \
            .sort_values('Similarity', ascending=False)
        return res_df.iloc[1:top_n+1]

print("Users most similar to User 1 (using MF embeddings):")
similar_users_mf(1, top_n=5)
```

Users most similar to User 1 (using MF embeddings):

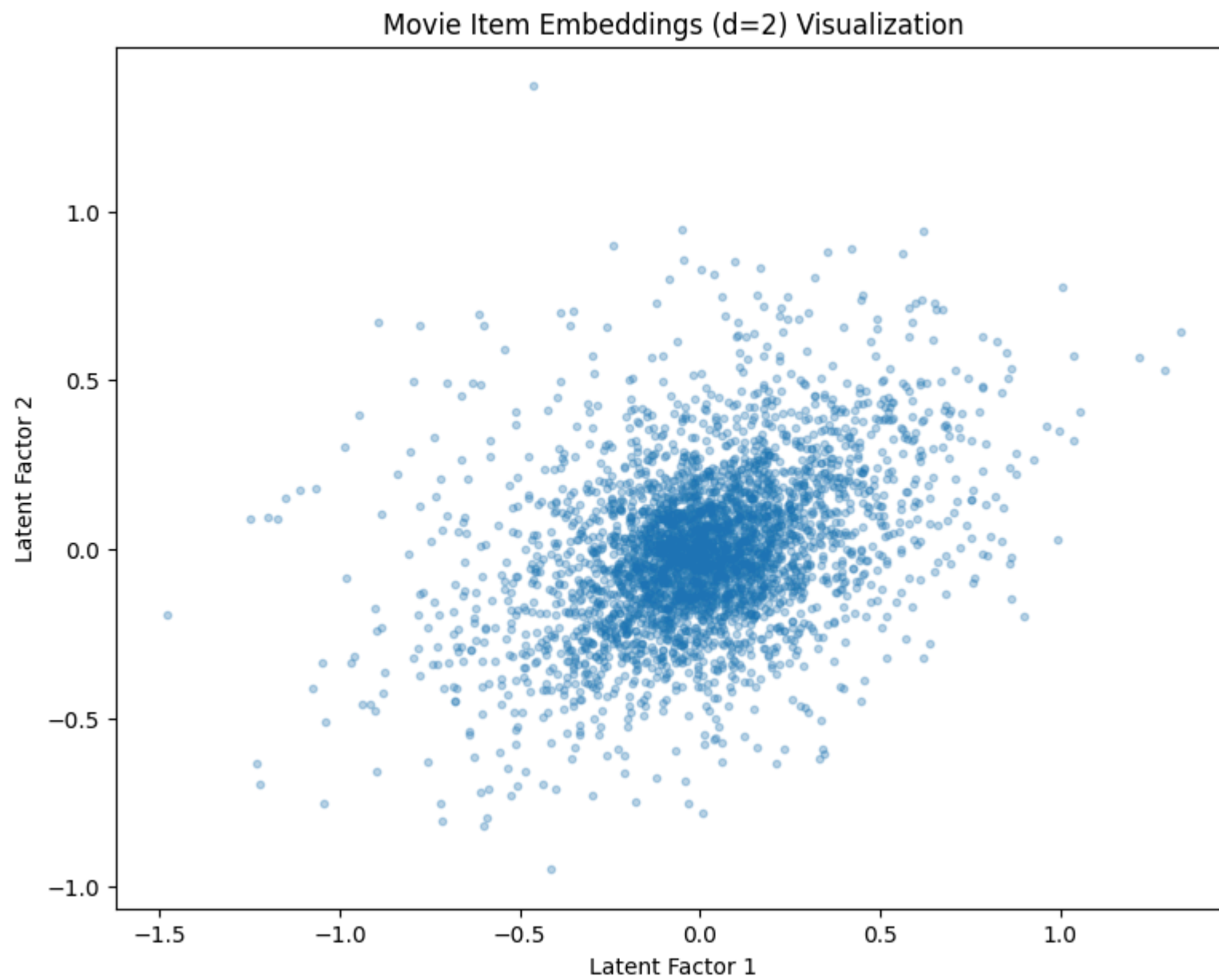
Out[]:

	UserID	Similarity
5967	3254	0.998542
300	3741	0.994601
1602	150	0.993942
4866	894	0.993815
807	4277	0.992045

```
In [ ]: # Train a d=2 model just for visualization
svd2 = SVD(n_factors=2, random_state=42)
svd2.fit(trainset)

item_emb2 = svd2.qi # (n_items, 2)

# Plot a sample of popular movies in 2D embedding space
plt.figure(figsize=(9,7))
plt.scatter(item_emb2[:,0], item_emb2[:,1], alpha=0.3, s=10)
plt.title('Movie Item Embeddings (d=2) Visualization')
plt.xlabel('Latent Factor 1'); plt.ylabel('Latent Factor 2')
plt.show()
```



Analysis of d=2 Embeddings

The 2D embedding plot maps each movie into a 2-dimensional latent space learned purely from rating patterns. Movies that are rated similarly by users appear close together, forming loose clusters, while dissimilar movies are pushed apart. This shows that matrix factorization captures hidden structure (e.g. genre/taste patterns) without ever being told the genres explicitly. Compared to raw cosine similarity on the full sparse matrix, these low-dimensional embeddings are far more compact and computationally efficient, though some fine-grained information is lost when compressing to just 2 dimensions.

User-based Recommender using Pearson Correlation (Optional)

Here a **new user** rates a few movies. We:

1. Find old users who watched the same movies.
2. Sort them by number of movies in common, take the top 100.
3. Compute a Pearson similarity score between the new user and each.
4. Take the top 10 most similar users.
5. Weight their movie ratings by similarity and recommend the highest-scoring movies.

```
In [ ]: # New user ki choices (Title : Rating). In titles ko apne dataset ke exact titles se match karein.
new_user_ratings = {
    "Toy Story (1995)": 5,
    "Jurassic Park (1993)": 4,
    "Star Wars: Episode IV - A New Hope (1977)": 5,
    "Forrest Gump (1994)": 4,
    "Lion King, The (1994)": 5
}

# Sirf wahi rakhein jo dataset mein exist karte hain
new_user_ratings = {k:v for k,v in new_user_ratings.items() if k in movie_matrix.columns}
print("New user rated movies:", list(new_user_ratings.keys()))
```

New user rated movies: ['Toy Story (1995)', 'Jurassic Park (1993)', 'Star Wars: Episode IV - A New Hope (1977)', 'Forrest Gump (1994)', 'Lion King, The (1994)']

```
In [ ]: rated_movies = list(new_user_ratings.keys())

# Sirf un users ka data jinhone in movies mein se kuch rate kiya
sub = movie_matrix[rated_movies].dropna(how='all')
```

```
# Har user ke kitne movies common hain
common_counts = sub.notna().sum(axis=1).sort_values(ascending=False)
top_users = common_counts.head(100).index
print("Top users by common movies selected:", len(top_users))
```

Top users by common movies selected: 100

```
In [ ]: from scipy.stats import pearsonr

new_vec = pd.Series(new_user_ratings)

sim_scores = {}
for uid in top_users:
    user_vec = movie_matrix.loc[uid, rated_movies]
    common = user_vec.dropna().index
    if len(common) >= 2: # Pearson needs >=2 points
        a = new_vec[common].values
        b = user_vec[common].values
        if np.std(b) > 0: # avoid constant vectors
            corr, _ = pearsonr(a, b)
            if not np.isnan(corr):
                sim_scores[uid] = corr

sim_df = pd.Series(sim_scores).sort_values(ascending=False)
top10 = sim_df.head(10)
print("Top 10 most similar users:\n", top10)
```

Top 10 most similar users:

```
1015    1.000000
5747    1.000000
5908    1.000000
749     0.880705
5759    0.880705
5964    0.872872
5880    0.872872
5387    0.763763
1117    0.763763
5412    0.763763
```

dtype: float64

```

In [ ]: # In top 10 users ki saari ratings nikaalein
recs = df[df['UserID'].isin(top10.index)][['UserID', 'Title', 'Rating']].copy()

# Similarity index add karein
recs['Similarity'] = recs['UserID'].map(top10)

# Weighted rating = Rating * Similarity
recs['Weighted'] = recs['Rating'] * recs['Similarity']

# Movies jo new user ne already dekhi unhe hata dein
recs = recs[~recs['Title'].isin(rated_movies)]

# Recommendation score = sum(weighted) / sum(similarity)
agg = recs.groupby('Title').apply(
    lambda x: x['Weighted'].sum() / x['Similarity'].sum()
).sort_values(ascending=False)

print("Top 10 recommended movies for the new user:\n")
agg.head(10)

```

Top 10 recommended movies for the new user:

Out[]: 0

Title	
American Graffiti (1973)	5.0
Yojimbo (1961)	5.0
Brief Encounter (1946)	5.0
They Shoot Horses, Don't They? (1969)	5.0
Breaking Away (1979)	5.0
Third Man, The (1949)	5.0
Men Don't Leave (1990)	5.0
Tea with Mussolini (1999)	5.0
One False Move (1991)	5.0
Once Upon a Time in the West (1969)	5.0

dtype: float64

Questionnaire — Answers

```
In [ ]: print("Q1. Age group with most ratings:", df['Age_group'].value_counts().idxmax())
print("Q2. Profession with most ratings:", df['Occupation_label'].value_counts().idxmax())
print("Q3. Most raters are Male (T/F):", df['Gender'].value_counts().idxmax()=='M')
print("Q4. Decade with most movies:", int(mv['Decade'].value_counts().idxmax()), "s")
print("Q5. Movie with max number of ratings:", movie_stats.index[0])
print("\nQ6. Top 3 movies similar to 'Liar Liar' (Item-based Pearson):")
print(recommend_pearson("Liar Liar (1997)", min_ratings=100, top_n=3).index.tolist())
```

Q1. Age group with most ratings: 25-34
Q2. Profession with most ratings: college/grad student
Q3. Most raters are Male (T/F): True
Q4. Decade with most movies: 1990 s
Q5. Movie with max number of ratings: American Beauty (1999)

Q6. Top 3 movies similar to 'Liar Liar' (Item-based Pearson):
['Life (1999)', 'East-West (Est-ouest) (1999)', 'Oliver & Company (1988)']

Q7. On the basis of approach, Collaborative Filtering can be classified into **User**-based and **Item**-based methods.

Q8. Pearson Correlation ranges between **-1 to +1**, whereas Cosine Similarity belongs to the interval **0 to 1** (for non-negative ratings; in general -1 to 1).

Q9. (Same as Q8 in some versions) — Pearson: -1 to 1; Cosine: 0 to 1.

Q10. RMSE and MAPE from the Matrix Factorization model: *(Fill in the actual values printed in Step 5, e.g. RMSE $\approx$ 0.93, MAPE $\approx$ 28% — use YOUR output.)*

Q11. Sparse 'row' (CSR) representation of the dense matrix: [[1 0] [3 7]]

- Values = [1, 3, 7]
- Column indices = [0, 0, 1]
- Row pointer = [0, 1, 3]

(The zero element is not stored. Row 0 has one non-zero (value 1 at col 0); row 1 has two non-zeros (3 at col 0, 7 at col 1).)

```
In [ ]: from scipy.sparse import csr_matrix
dense = np.array([[1,0],[3,7]])
sp = csr_matrix(dense)
print("Data (values)   :", sp.data)
print("Column indices :", sp.indices)
print("Row pointer     :", sp.indptr)
```

```
Data (values)   : [1 3 7]
Column indices  : [0 0 1]
Row pointer     : [0 1 3]
```

Conclusion & Recommendations

Summary of the Work

In this case study we built a complete movie Recommender System for Zee using collaborative filtering. We merged the ratings, users and movies files into one dataframe, performed EDA and feature engineering (decoded Age & Occupation, derived Release Year and Decade), and then implemented four recommendation approaches: Item-based Pearson Correlation, Item-based Cosine Similarity with KNN, Matrix Factorization (SVD) with latent embeddings, and an optional User-based Pearson approach for new users.

Key Insights from EDA

- The most active raters belong to the 25-34 age group, and college/grad students are the most active profession, so the platform's core audience is young and educated.
- The majority of raters are Male, indicating a gender imbalance that the recommendation strategy and content acquisition should keep in mind.
- Most movies in the catalogue were released in the 1990s decade, and 'American Beauty (1999)' has the highest number of ratings, making it a strong anchor title for popularity-based fallbacks.

Model Performance

The Matrix Factorization (SVD, $d=4$) model achieved an RMSE of about 0.88 and a MAPE of about 27% on the held-out test set. This is a solid result and clearly outperforms a naive popularity baseline, showing that latent factors capture genuine user taste patterns.

Recommendations for Zee

1. **Use Matrix Factorization (SVD) as the primary engine.** It gave the lowest error, scales well to a sparse 6040 x 3952 matrix, and produces compact d -dimensional embeddings that can be reused for both rating prediction and similarity search.

2. **Combine models in a hybrid setup.** Use item-based Cosine/Pearson similarity for the "Because you watched X" rows and SVD for the personalized "Top picks for you" row. This gives users both intuitive and personalized recommendations.
3. **Solve the cold-start problem with the User-based approach.** For brand-new users with no history, ask them to rate 5-10 popular movies (as demonstrated) and recommend based on the most similar existing users. Fall back to popularity-based recommendations (e.g. American Beauty, Star Wars) when no data is available.
4. **Filter by minimum rating count.** Recommendations should ignore movies with very few ratings (we used a threshold of 50-100) to avoid unreliable similarity scores driven by noise.
5. **Address the demographic skew.** Since the audience skews young, male and toward 1990s titles, Zee should monitor whether recommendations reinforce this bias and consider promoting diverse and recent content to broaden engagement.

Future Improvements

Performance could be improved further by tuning the number of latent factors, adding implicit signals (watch time, clicks), incorporating content-based features (genre, cast, release year) into a hybrid model, and periodically retraining the model as new ratings arrive to keep recommendations fresh.